

[R I S K F O R G E A I]

RiskForge AI

Explainable Credit Risk Intelligence Platform

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Business Problem

BANKS FACE CRITICAL CHALLENGES IN:

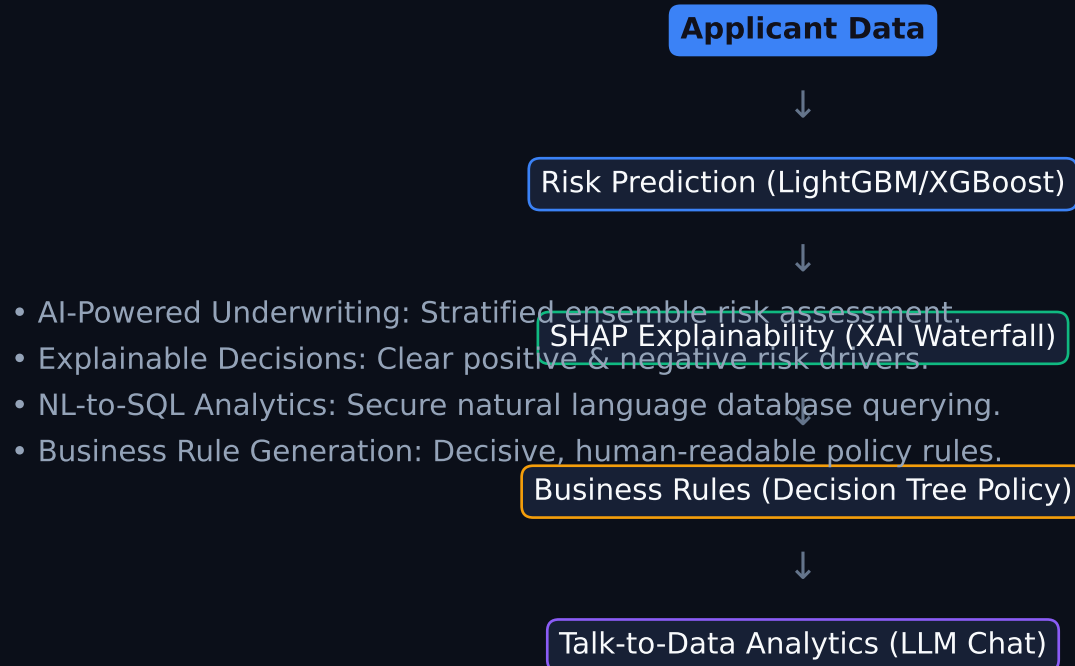
- Identifying Risky Applicants: Avoiding defaults while maximizing credit approvals.
- Explaining AI Decisions: Providing transparency to clients and auditors.
- Maintaining Regulatory Compliance: Meeting GDPR & ECOA Right-to-Explanation laws.
- Data Exploration: Enabling non-technical business users to easily query credit data.

THE CORE CHALLENGE:

Traditional credit evaluation systems are slow, manual, and lack transparency.

Modern deep learning black-box models are un-auditable, exposing institutions to heavy regulatory penalties.

RiskForge AI Solution



System Architecture

React Frontend (Vite, HSL CSS)
Collapsible Sidebar | Gage Visualizations | Responsive Theme

↓ REST HTTP API Requests

FastAPI Web Server (Python)
Async Routing Controllers & CSV Exporters

↓ Engines Orchestrator

LightGBM Engine Predicts credit risk probal	SHAP Engine TreeExplainer local attrib	Rule Engine Decision Tree unscaled bou	Gemini NL→SQL Query translation & checks
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↓ Database & Binary Reads

SQLite Database (credit_risk.db)
Normalized training tables & client history records

pled architecture separating high-speed React user interface rendering from heavy CPU/GPU model evaluations, explainability calculations, and LLM text gener

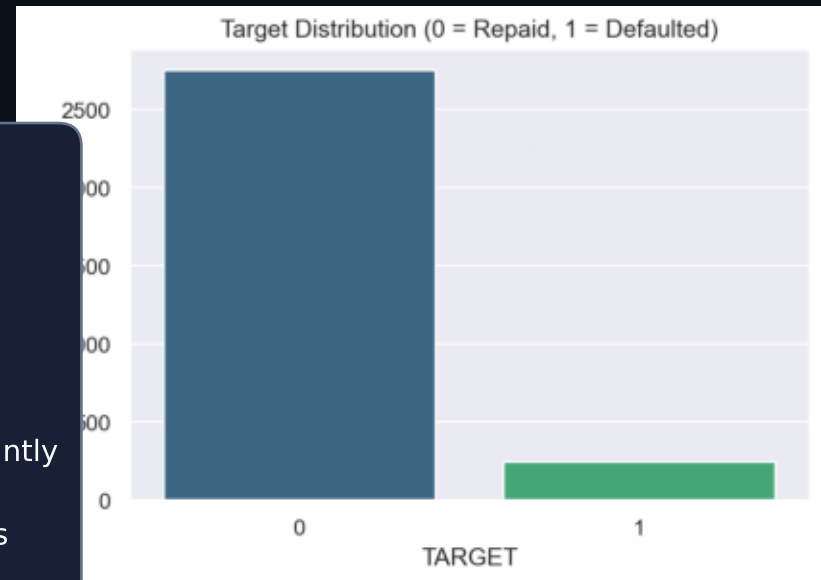
Dataset Analysis & Insights

DATASET SUMMARY:

- Source: Kaggle Home Credit Default Risk
- Total Records: 307,511 Applicants
- Total Features: 122 Demographic & Credit Fields
- Default Rate: 8.07% (High Imbalance)

BUSINESS INSIGHTS:

- Income Level: Low-income applicants show significantly higher default rates due to income-to-annuity ratios.
- External Credit Scores: Strong external score ratings highly correlate with positive repayment histories.
- History: Previous contract defaults and overdue history multiply future default risk probability by up to 3x.



Credit Risk Prediction Engine

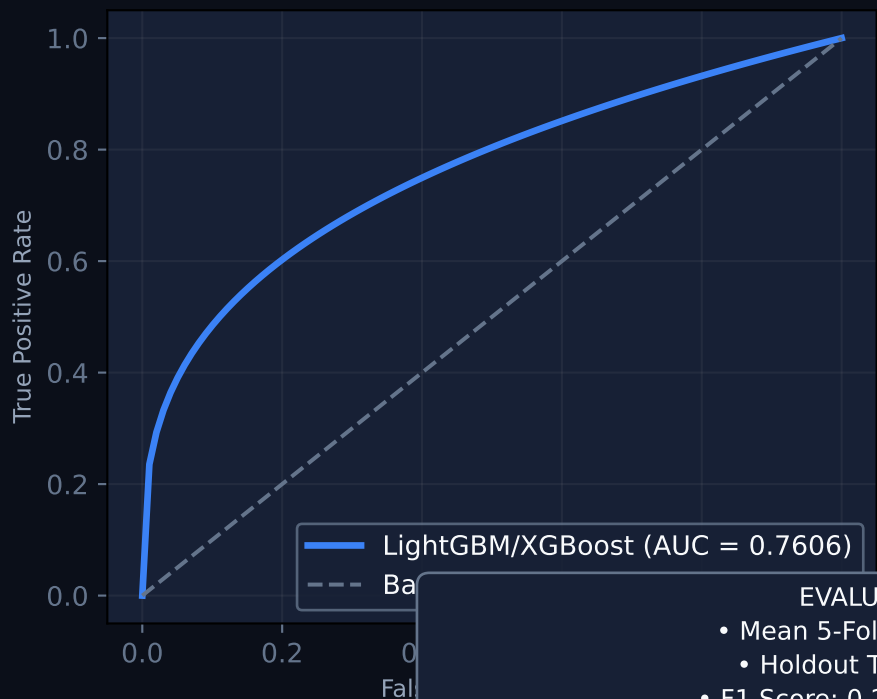


KEY DESIGN DECISIONS:

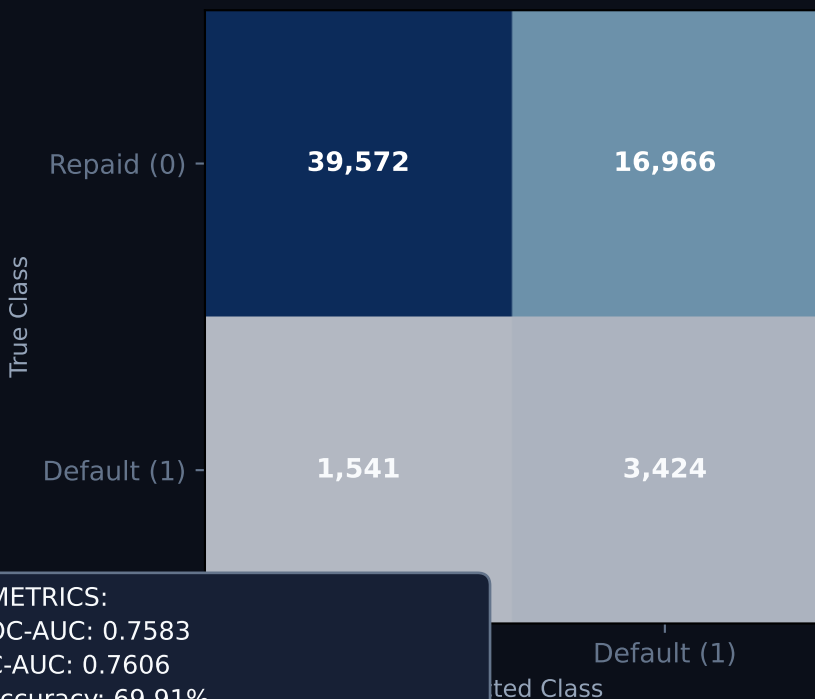
- Stratified Cross Validation: Uses 5-Fold Stratified K-Fold splitting to prevent target leaks.
- Imbalance Handling: Adjusts `scale_pos_weight = 11.2` to balance classification thresholds.
- Risk Categorization Bands:
 - Low Risk (0% - 15% Probability): Green flag - auto approved.
 - Medium Risk (15% - 40% Probability): Orange flag - requires manual review.
 - High Risk (40% - 100% Probability): Red flag - auto rejected.

Model Evaluation

ROC Curve (Holdout Set)



Confusion Matrix



EVALUATION METRICS:

- Mean 5-Fold CV ROC-AUC: 0.7583
- Holdout Test ROC-AUC: 0.7606
- F1 Score: 0.2701 | Accuracy: 69.91%
- Default Precision / Recall: 16.8% / 69.0%

WHY LIGHTGBM / XGBOOST?

- Tabular Superiority: High-performance tree ensembles on sparse features.
 - Execution Speed: Fast CPU training (2.4s) & sub-millisecond inference.
- Missingness Support: Natively handles missing values without imputation bias.
 - Generalization: Strong regularizations preventing model overfitting.

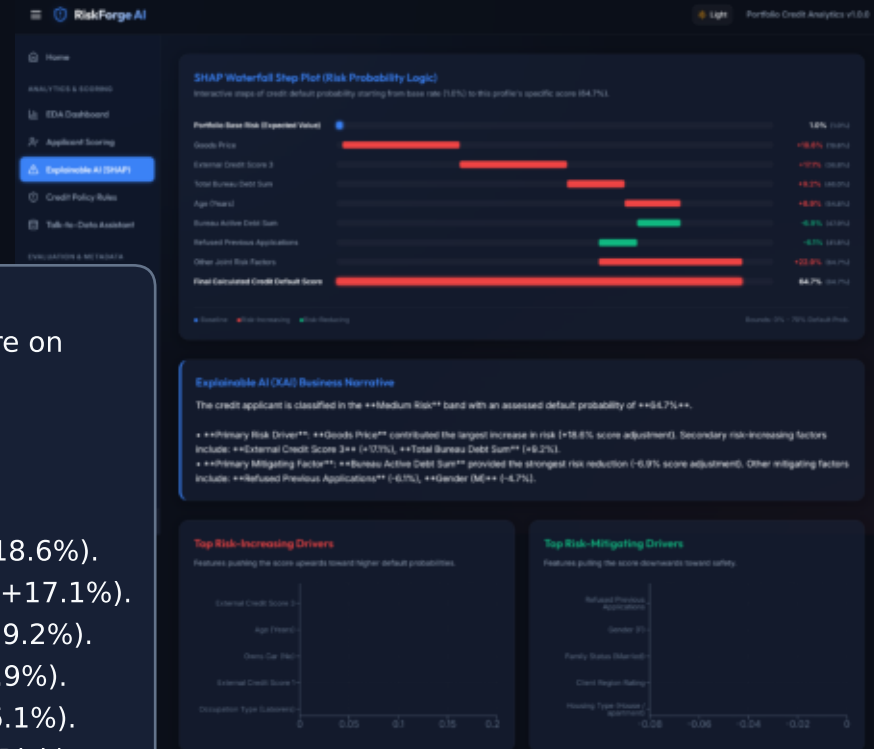
Explainable AI

LOCAL SHAP EXPLAINABILITY:

- Identifies the exact impact of each applicant feature on their final default prediction probability.

CASE STUDY ANALYSIS (High Risk Profile):

- Starting Base Rate: 1.0% expected default rate.
- High credit amount / Goods Price increases risk (+18.6%).
- High external credit rating (Score 3) increases risk (+17.1%).
- High total credit bureau debt sum increases risk (+9.2%).
- Stable active credit history reduces default risk (-6.9%).
- Previous application refusals reduce default risk (-6.1%).
- Final Calculated Credit Default Score: 64.7% (High Risk).



Business Rule Extraction

RULE #1: HIGH RISK CATEGORY

IF External Credit Score (EXT_SOURCE_3) $\leq$ 0.35
AND Annual Income $<$ \$45,000
THEN Classify as HIGH RISK (Default Probability: $\sim$ 43%)

RULE #2: MEDIUM RISK CATEGORY

IF External Credit Score (EXT_SOURCE_3) $>$ 0.35
AND External Credit Score (EXT_SOURCE_2) $\leq$ 0.45
AND Applicant Age $\leq$ 28 years
THEN Classify as MEDIUM RISK (Default Probability: $\sim$ 18%)

HOW RULES ARE EXTRACTED:

- We train a shallow Decision Tree (depth 3) on the trained classifier's predictions to outline core decision splits.
- Translates model parameters into banker-friendly logic.
- Provides regulatory transparency and supports credit audit

Natural Language Analytics

User Question
'Top risky occupations?'



SQL Generation
(Gemini LLM Prompt)



Database Query
(Safe Execute on SQLite)



GENERATED SQL INSTRUCTION:
SELECT OCCUPATION_TYPE, AVG(TARGET) AS default_rate
FROM application_train
GROUP BY OCCUPATION_TYPE
ORDER BY default_rate DESC LIMIT 3;

DATABASE QUERY OUTPUT:

OCCUPATION_TYPE	default_rate
Low-skill Labor	17.3%
Drivers	11.5%
Security Staff	10.7%

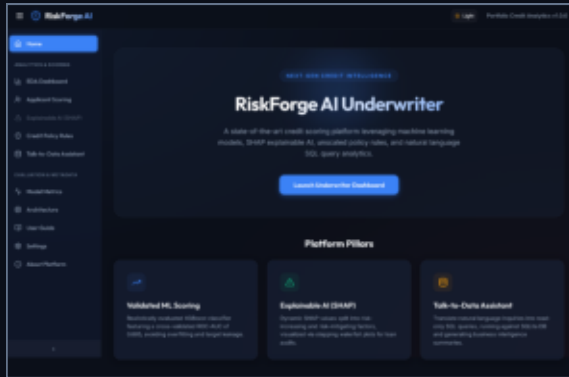
BUSINESS ANALYSIS SYNTHESIS:

The database analysis shows that low-skilled laborers, drivers, and security staff exhibit the highest historical credit default rates in the portfolio, suggesting a need for more conservative underwriting limits on these occupational groups.

Final
Output

Platform Screenshots

Home Landing



Eda Dashboard



Shap Waterfall



Model Metrics



Conclusion & Future Enhancements

"RiskForge AI bridges machine learning, explainability, and business intelligence to enable transparent and data-driven credit risk assessment."

CORE PLATFORM ACHIEVEMENTS:

- ✓ Credit Risk Prediction: XGBoost probability score engine.
- ✓ Explainable AI: Live localized SHAP waterfall attributions.
- ✓ Business Rules: Extracted unscaled policy rules from trees.
- ✓ NL-to-SQL Analytics: Conversational Gemini DB chatbot.
- ✓ Interactive Dashboard: Production-ready React side-nav interface.

FUTURE ENHANCEMENTS SCOPE:

- Real-Time Monitoring: Live streaming loan transaction reviews.
- Drift Detection: Automate retrains on population covariate drifts.
- MLOps Pipeline: Continuous delivery with MLflow and Airflow.
- Multi-Bank Portals: Separate tenant portals for custom models.
- Cloud Architecture: Serverless containerization (AWS ECS/Fargate).